

Prediction of Appendicitis Diagnosis Using Machine Learning with Clinical and Laboratory Data

Introduction.	Pediatric appendicitis is a common surgical emergency requiring timely and accurate diagnosis to prevent serious complications like perforation. However, diagnosis is often difficult due to nonspecific symptoms and overlapping clinical signs, especially in younger children. Traditional methods—clinical evaluation, lab tests, and imaging—can be variable and subjective, leading to misdiagnosis or unnecessary surgeries. Machine learning can enhance diagnosis by integrating diverse data sources to provide rapid, objective, and accurate risk assessments. Using the comprehensive Regensburg Pediatric Appendicitis dataset, ML models have the potential to improve clinical decision-making and patient outcomes by identifying patterns beyond conventional approaches.
Objectives.	- To develop and evaluate a machine learning model for predicting the presence of appendicitis in pediatric patients. - To identify the most important predictors influencing appendicitis diagnosis through feature importance analysis.
Dataset Description.	- Data Source: https://archive.ics.uci.edu/dataset/938/regensburg+pediatric+appendicitis - Sample: 782 pediatric patients (2016–2021). - Variables: Clinical, laboratory and ultrasound data, a total of 53 variables (Variable descriptions are available on the data source link) - Target Variable: Confirmed diagnosis of appendicitis
Significance of the study.	Appendicitis is a common emergency in children that needs quick and accurate diagnosis. However, it can be hard to tell if a child really has appendicitis because the symptoms are similar to other illnesses. If doctors make a wrong decision, children might have unnecessary surgery or face dangerous delays in treatment. This study aims to use machine learning to help doctors predict if a child has appendicitis more accurately. By analyzing the detailed medical information in the Regensburg dataset, the model can give doctors quick and reliable advice. This can lead to better care, fewer unnecessary surgeries, and safer, faster treatment for children with appendicitis.
Methodology	- Data Preprocessing: Handling missing data, Variable selection, Data splitting - Model Development: Various Statistical and Machine Learning models to be explored (e.g., logistic regression, decision trees, random forest, XGBoost) - Model Evaluation: Evaluate models using performance metrics (accuracy, precision, recall, f1-score, AUC-ROC) - Feature Importance Analysis: Assess the relative contribution of each feature to model predictions and use this analysis to refine feature selection.
Reference	https://arxiv.org/abs/2302.14460v2 https://doi.org/10.3389/fped.2021.662183 https://pmc.ncbi.nlm.nih.gov/articles/PMC8116489/ https://doi.org/10.3390/jcm12041168